

Cheng-Yen (Chris) Yang

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I am a Ph.D. candidate in the Department of Electrical Computer Engineering at the University of Washington planning to start working in late 2025. My research interests lie around **Multi-Object Tracking (Single-View, Multi-View, Cross-View) in complex scenes (Underwater, Drone, Multi-Camera System)**.

Educations

University of Washington

Ph.D. in Electrical and Computer Engineering - GPA: 3.85

M.S. in Electrical and Computer Engineering - GPA: 3.87

- Faculty Advisor: Professor Jenq-Neng Hwang

National Taiwan University

B.S. in Electrical Engineering - GPA: 3.78

- 2018 The Presidential Award (Top 1% in Department of Electrical Engineering)

Seattle, United States

Mar. 2021 - Dec. 2025 (exp)

Sep. 2019 - Mar. 2021

Taipei, Taiwan

Sep. 2014 - Jan. 2019

Intern Experiences

Deep Learning Engineer Intern

Nvidia

- Worked with Dr. Thomas Tang and Dr. Yizhou Wang from the Metropolis Microservices Team to developing an online Multi-Camera Multi-Object Tracking method.

Jun. 2024 - Nov. 2024

Santa Clara, United States

Research Scientist Intern

Meta Reality Labs

- Worked with Dr. Sophia Fan, Dr. Kunpeng Li and Dr. Dimitri Model from the XR Eye team.
- Work on leveraging domain adaptation to enhance gaze estimation performance.

Jun. 2022 - Sep. 2022

Burlingame, United States

Applied Scientist Intern

Amazon.com

- Mentored by Dr. Jiajia Luo and Dr.Cheng-Hao Kuo from the Visual Perception team.
- Work on leveraging in-the-wild 2D annotations to enhance the 3D pose estimation performance on the cross-dataset scenarios, the work is accepted by WACV 2023.

Jun. 2021 - Sep. 2021

Bellevue, United States

Honors & Awards

- 2024 **1st Place**, Multi-Modal Visual Pattern Recognition Challenge: Multi-Modal Tracking Track (ICPR2024)
- 2024 **1st Place**, 2nd MaCVi Workshop Challenge: UAV-based Multi-Object Tracking (WACV2024)
- 2024 **1st Place**, 2nd MaCVi Workshop Challenge: USV-based Multi-Object Tracking (WACV2024)
- 2023 **1st Place**, 2023 AI CITY Challenge - Track 1: Multi-Camera People Tracking (CVPR2023)
- 2022 **3rd Place**, 5th MMSports - DeepSportRadar Player Reidentification Challenge (MM2022)
- 2021 **1st Place**, 6th Workshop on Benchmarking Multi-Target Tracking - Video Track (ICCV2021)
- 2021 **3rd Place**, 2021 Multi-camera Multiple People Tracking Challenge - Camera-View Track (ICCV)
- 2018 **The Presidential Award (Top 1%)**, National Taiwan University
- 2017 **National Technology and Research Scholarship**, CTCI Foundation, Taiwan

Research Experiences

Information Processing Lab, University of Washington

Research Assistant (Advisor: Prof. Jenq-Neng Hwang)

- Received funding from ETRI (South Korea), NIWA (New Zealand), and NOAA (US) to work on various projects on leveraging deep learning to solve real-world problems.
- Led and organized the team to participate in the 2024 MaCVi Challenge and won 1st place in two tracks: UAV-based Multi-Object Tracking and USV-based Multi-Object Tracking.
- Led and organized the team to participate in the 2023 AI CITY Challenge - Track 1: Multi-Camera People Tracking and won 1st place.

Aug. 2020 - present

Seattle, United States

Selected Publications

[1] ***SAMURAI: Adapting Segment Anything Model for Zero-Shot Visual Tracking with Motion-Aware Memory***

C.-Y. Yang, H.-W. Huang, W. Chai, Z. Jiang, J.-N. Hwang

ArXiv 2024.

[2] ***Exploring Learning-based Motion Models in Multi-Object Tracking***

H.-W. Huang, C.-Y. Yang, W. Chai, Z. Jiang, J.-N. Hwang

Accepted by International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2025.

[3] ***A Density-Guided Temporal Attention Transformer for Indiscernible Object Counting in Underwater Videos***

C.-Y. Yang, H.-W. Huang, Z. Jiang, H. Wang, F. Wallace, J.-N. Hwang.

Accepted by International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024.

[4] ***Back to Optimization: Diffusion-based Zero-Shot 3D Human Pose Estimation***

Z. Jiang, Z. Zhou, L. Li, W. Chai, C.-Y. Yang, J.-N. Hwang

Accepted by Winter Conference on Applications of Computer Vision (WACV) 2024.

[5] ***Multi-Object Tracking by Iteratively Associating Detections with Uniform Appearance for Trawl-Based Fishing Bycatch Monitoring***

C.-Y. Yang, A. Y. S. Tan, M. J. Underwood, C. Bodie, Z. Jiang, S. George, K. Warr, J.-N. Hwang, E. Jones

Accepted by International Conference on Image Processing (ICIP) 2023.

[6] ***CameraPose: Weakly-Supervised 3D Human Pose Estimation from a Single Image with Unlabeled in the Wild Data***

C.-Y. Yang, J. Luo, X. Lu, Y. Sun, Z. Jiang, J.-N. Hwang, C.-H. Kuo

Accepted by Winter Conference on Applications of Computer Vision (WACV) 2023.

[7] ***Unsupervised Domain Adaptation Learning for Infant Pose Recognition with Synthetic Data***

C.-Y. Yang, Z. Jiang, S.-Y. Gu, J.-N. Hwang, J.-H. Yoo

Accepted by International Conference of Multimedia and Expo (ICME) 2022.

Workshop Publications

[1] ***An online approach and evaluation method for tracking people across cameras in extremely long video sequence***

C.-Y. Yang, H.-W. Huang, Z. Jiang, P.-K. Kim, K. Lee, K. Kim, I.-S. Jang, C.-I. Huang, J.-N. Hwang

Accepted by the 7th AI City Challenge Workshop (in CVPR2023).

[2] ***Sea You Later: Metadata-Guided Long-Term Re-Identification for UAV-Based Multi-Object Tracking***

C.-Y. Yang, H.-W. Huang, Z. Jiang, H.-C. Kuo, J. Mei, C.-I. Huang, J.-N. Hwang

Accepted by 2nd Workshop of Maritime Computer Vision (in WACV 2024). **Winning method of the UAV Multi-Object Tracking Challenge.**

[3] ***Iterative Scale-Up ExpansionIoU and Deep Features Association for Multi-Object Tracking in Sports***

H.-W. Huang, C.-Y. Yang, J. Sun, P.-K. Kim, K.-J. Kim, K. Lee, C.-I. Huang, J.-N. Hwang

Accepted by Workshop of Real-World Surveillance: Applications and Challenges (in WACV2024).

[4] ***Enhancing Multi-Camera People Tracking with Anchor-Guided Clustering and Spatio-Temporal Consistency ID Re-Assignment***

C.-Y. Yang, H.-W. Huang, Z. Jiang, P.-K. Kim, K. Lee, K. Kim, S. Ramkumar, C. Mullapudi, I.-S. Jang, C.-I. Huang, J.-N. Hwang

Accepted by the 7th AI City Challenge Workshop (in CVPR2023). **Winning method of Track 1: Multi-Camera People Tracking.**

[5] ***U3D-MOLTS: Unified 3D Monocular Object Localization, Tracking and Segmentation***

H. Zhang, Y. Wang, Z. Jiang, C.-Y. Yang, J. Mei, J. Cai, J.-N. Hwang

Appeared in 6th Workshop on Benchmarking Multi-Target Tracking (in ICCV2021). **Winning method of Video Track: KITTI-STEP.**

[6] ***HVPS: A Human Video Panoptic Segmentation Framework***

Y. Wang, H. Zhang, Z. Jiang, J. Mei, C.-Y. Yang, J. Cai, J.-N. Hwang

Appeared in 6th Workshop on Benchmarking Multi-Target Tracking (in ICCV2021). **Winning method of Video Track: MOTChallenge-STEP.**

Reviewing Experiences

Program	UW ECE MS Admissions Triage Committee ('20-'21, '21-'22, '22-'23)
Conference	IEEE/CVF WACV, IEEE ICIP, IEEE ICME, IEEE ICCV
Journals	IEEE T-ITS, IEEE TCSVT, Electronics